# Namrata Nadagouda

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## OVERVIEW

- Research on developing methods for learning data efficient models based on active learning, human-in-the-loop learning and learning from human feedback/preferences
- Worked on applications such as preference learning, control parameter tuning, metric learning, and classification, and worked with images and sequential data
- Have a well rounded knowledge of ML theory and practice including both traditional ML and deep learning methods
- Experience with Python, MATLAB, PyTorch, SQL, Labelbox, Azure, AWS

## EDUCATION

**Ph.D. Electrical & Computer Engineering** *December 2026 (Expected)*
Machine Learning and Digital Signal Processing
*Georgia Institute of Technology, Atlanta, GA*
*Advisor*: Prof. Mark Davenport
*Thesis*: "Building data efficient models using active learning and similarity comparisons"

**M.S. Electrical & Computer Engineering** December 2020
Machine Learning and Digital Signal Processing
*Georgia Institute of Technology, Atlanta, GA*

**B.Tech. Electrical & Electronics Engineering** May 2017
Digital Signal Processing and Digital System Design
*National Institute of Technology Karnataka, Surathkal, India*

## PUBLICATIONS

**N. Nadagouda**, N. Ahad, M. Tucker and M. Davenport, "Active query synthesis for preference learning", *Under review.* Also presented at Women in Machine Learning (WiML) Symposium, co-located with *International Conference on Machine Learning (ICML), July 2026.*

**N. Nadagouda**, A. Xu and M. Davenport, "Active metric learning and classification using similarity queries", in *Uncertainty in Artificial Intelligence (UAI)*, August 2023. Also presented at Human in the Loop Learning Workshop, *Neural Information Processing Systems (NeurIPS), December 2022.*

A. McRae, A. Xu, J. Jin, **N. Nadagouda**, N. Ahad, P. Guan, S. Karnik and M. Davenport, "Delta Distancing: A Lifting Approach to Localizing Items From User Comparisons", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), May 2022.*

**N. Nadagouda** and M. Davenport, "Switched Hawkes Processes", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), June 2021.*

G. Canal, M. Connor, J. Jin, **N. Nadagouda**, M. O'Shaughnessy, C. Rozell and M. Davenport, "The Picasso Algorithm for Bayesian Localization Via Paired Comparisons in a Union of Subspaces Model", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), May 2020.*

**ABSTRACTS**

N. Ahad, **N. Nadagouda**, E. Dyer and M. Davenport, "Active learning for time instant classification", at Data-centric Machine Learning Research Workshop, *International Conference on Machine Learning (ICML), July 2023.*

Y. Teng, A. Mamuye, E. Mo, K. Zhu, R. Walker, **N. Nadagouda** and M. Davenport, "Range-Only Simultaneous Localization and Mapping using Paired Comparisons", at *IEEE Annual Conf. on RFID, April 2021.*

**N. Nadagouda** and M. Davenport, "Switched Hawkes Processes", at *Workshop on Recent Developments on Mathematical/Statistical approaches in Data Science (MSDAS), June 2019.*

**ARTICLES**

**N. Nadagouda**, "Journey of a researcher: Finding pleasure in the pathless woods", *American Ceramic Society Bulletin*, Student Perspectives, June/July 2020.

**WORK EXPERIENCE**

**Applied Scientist Intern** Summer 2024
*ML Science Team, Samsara, San Francisco, CA*

- Worked on active learning based traffic sign detection model using trip still images from dashcams
- Built the infrastructure to actively select images for labeling using a pre-trained distilled model and fine-tune the model using the labels obtained
- Also, involved data preparation which comprised of text prompt based collection of data using a CLIP based similarity search tool and labeling using Labelbox
- Conducted initial experiments with varying batch sizes of acquired data and the findings were handed off to the team for further investigation

**Intern** Summer 2018
*Hedge Fund Start-up, Atlanta, GA*

- Worked on data management and data pre-processing of stock trade data stored in SQL databases.
- Also, involved performance evaluation of trading algorithms on Microsoft Azure platform

**Research Intern, Microarchitecture Research Lab** Fall 2016
*Intel India - Intel Labs, Bangalore, India*

**Research Intern** Summer 2016
*Department of Electrical Communication Engineering*
*Indian Institute of Science, Bangalore, India*

**TALKS**

**Active query synthesis for preference learning**
*ML@GT Student Conference, Georgia Tech, Atlanta, GA* 2025

**Active query synthesis for preference learning**
*AWM Workshop at SIAM Annual Meeting, Montreal, Canada* 2025

**Active learning for traffic sign detection**
*Samsara, San Francisco, CA* 2024

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| | **Building data efficient models using active learning and similarity comparisons**<br>*Thesis presentation at Georgia Tech, Atlanta, GA* | 2022 |
| | **Switched Hawkes Processes**<br>*Student Seminar at Georgia Tech, Atlanta, GA* | 2019 |
| **AWARDS** | 2nd place, Presentations<br>*ML@GT Student Conference*<br>Georgia Tech | September 2025 |
| **Registration and Travel Awards** | AWM Workshop at SIAM Annual Meeting<br>*Travel Funding* | July 2025 |
| | NeurIPS<br>*Conference Volunteer Financial Assistance*<br>*WiML Travel Funding* | December 2023 |
| | Women in Data Science and Mathematics<br>Institute for Pure and Applied Mathematics<br>*Workshop Financial Assistance*<br>*GT Career Development Funding* | August 2023 |
| | UAI<br>*Conference Funding* | August 2023 |
| | ICML<br>*GT SGA Conference Funding*<br>*CoE Professional Development Funding*<br>*WiML Funding* | August 2023 |
| | NeurIPS<br>*GT SGA Conference Funding*<br>*CoE Professional Development Funding* | December 2022 |
| | Deep Learning Theory Workshop and Summer School<br>Simons Institute<br>*Workshop Financial Assistance* | August 2022 |
| | The Mathematics of Machine Learning<br>Women and Math Program<br>Institute for Advanced Study<br>*Workshop Financial Assistance* | May 2022 |
| | Recent Developments on Mathematical/Statistical approaches in Data Science<br>University of Texas, Dallas<br>*Workshop Financial Assistance* | June 2019 |
| **Hackathon** | Winner, Technical Track, Hacklytics<br>Data Science at Georgia Tech | February 2019 |
| **Academic Awards and Scholarships** | NITK Institute Gold Medal<br>1986 Batch Gold Medal | 2017<br>2017 |

Prof. M. R. Shenoy Memorial Prize 2017
Prof. K. M. Hebbar Gold Medal 2017
NITK Surathkal Merit Scholarship 2013 - 2017

## SERVICE

**Reviewer**, NeurIPS Conference 2026
**Reviewer**, WiML Workshop at NeurIPS 2025
**Judge**, AWM Poster Session at SIAM Annual Meeting 2025
**Mentor**, AWM Workshop at SIAM Annual Meeting 2025
**Reviewer**, Workshop Proposals for NeurIPS 2025
**Reviewer**, WiML Workshop at NeurIPS 2024
**Reviewer/Volunteer**, WiML workshop at NeurIPS 2023
**Volunteer**, NeurIPS Conference 2023
**Volunteer**, UAI Conference 2023
**Volunteer**, WiML Workshop at ICML 2023
**Member**, GT Mural Team 2022
**Reviewer**, AIStats Conference 2021
**Reviewer**, GT President's Undergraduate Research Award 2021 - 2022
**Panelist**, GT ECE ORS Graduate Panel 2020
**Volunteer**, GT ECE Prospective PhD student visit 2019 - 2022
**Teaching Volunteer**, Shiksha, ACM NITK Student Chapter 2016
**Student Representative**, NITK Student Council 2013 - 2014

## TEACHING EXPERIENCE

**Mentor for undergraduate students**
Yue Teng 2020 - 2022
Amran Mamuye, Eunsan Mo, Kerui Zhu, Robert Walker 2020 - 2021
Guided the above students to work on a research project focused on simultaneous localization and mapping using paired comparisons of distances.

**Graduate Teaching Assistant** Spring 2021
Georgia Tech ECE 6270 - Convex Optimization

**Teaching Assistant** June 2019
Hands-on-Tech Georgia Tech Day Camp - Machine Learning

**Graduate Teaching Assistant** Spring 2019
Georgia Tech CS 4641 - Machine Learning

**Graduate Teaching Assistant** Fall 2018
Georgia Tech ECE 8843/ISYE 8843/CS 8803/BMED 8813 -
Mathematical Foundations of Machine Learning

## RESEARCH EXPERIENCE

**Control parameter tuning using active synthesis** Fall 2025 - present
Working on applying active preference learning to optimize subjective controller gains for mobile robots. The framework dynamically synthesizes continuous-space queries to compare different trajectory parameters, asking users which control behavior they prefer. By translating this qualitative human feedback into a structured search, the method iteratively identifies optimal control parameters with minimal queries.

**Active query synthesis for preference learning** Fall 2022 - present
Working on improving preference learning with a confidence-aware response model that selects highly reliable queries by targeting the ideal geometric distance between compared items. To eliminate the computational bottlenecks of standard methods, the active query synthesis framework directly generates optimal queries within a continuous space by maximizing a mutual information-based objective. For scenarios

restricted to finite data pools, the method employs tailored approximation strategies to efficiently map these continuous ideal queries to the best available real-world pairs.

**Active learning for time series data** Summer 2023
Working on developing a method for classifying individual instants of time series data. The data consists of features which repeat at regular intervals and the existence of these correlations poses a unique challenge for active label selection.

**Unified framework for active learning** Fall 2019 - Spring 2023
Developed a unified query framework for active learning based on nearest neighbor queries. This method can be applied to any problem which involves learning a representation of the dataset that reflects the underlying similarity. Demonstrated the performance of the method for active metric learning and active image classification using deep neural networks.

**Active similarity learning and manifold graphs** Fall 2019 - Spring 2023
Implemented active image classification strategies for semi-supervised classification on CIFAR-100 and DomainNet datasets. This project was funded by the *DARPA LwLL – Learning with Less labels* program.

**Preference learning** Fall 2019 - Summer 2022
The problem consists of estimating a user's preferences over a set of items. We use the *ideal point* model to localize a user in an embedding of items. Worked on a variety of problems involving localizing new items and users.

**Switched Hawkes Processes** Fall 2018 - Summer 2019
Developed the Switched Hawkes Process which can be used to model systems in which the parameters of the process dynamically change depending on some (known) external state. We propose a simple maximum likelihood estimation approach and apply our model to a real-world traffic sensor dataset to study traffic patterns during different configurations of the traffic lights at an intersection.

## WORKSHOPS ATTENDED

Women in Data Science and Mathematics — August 2023
Institute for Pure and Applied Mathematics
*University of California, Los Angeles*

Deep Learning Theory Workshop and Summer School — August 2022
*Simons Institute, University of California, Berkeley*

The Mathematics of Machine Learning — May 2022
Women and Math Program
*Institute for Advanced Study, Princeton, NJ*

Algorithmic Learning Theory — March 2022
Mentoring Workshop
*Online*

Foundation of Data Science Summer School — August 2019
*Georgia Institute of Technology, Atlanta, GA*

Recent Developments on Mathematical/Statistical approaches in Data Science — June 2019
*University of Texas at Dallas, Richardson, TX*